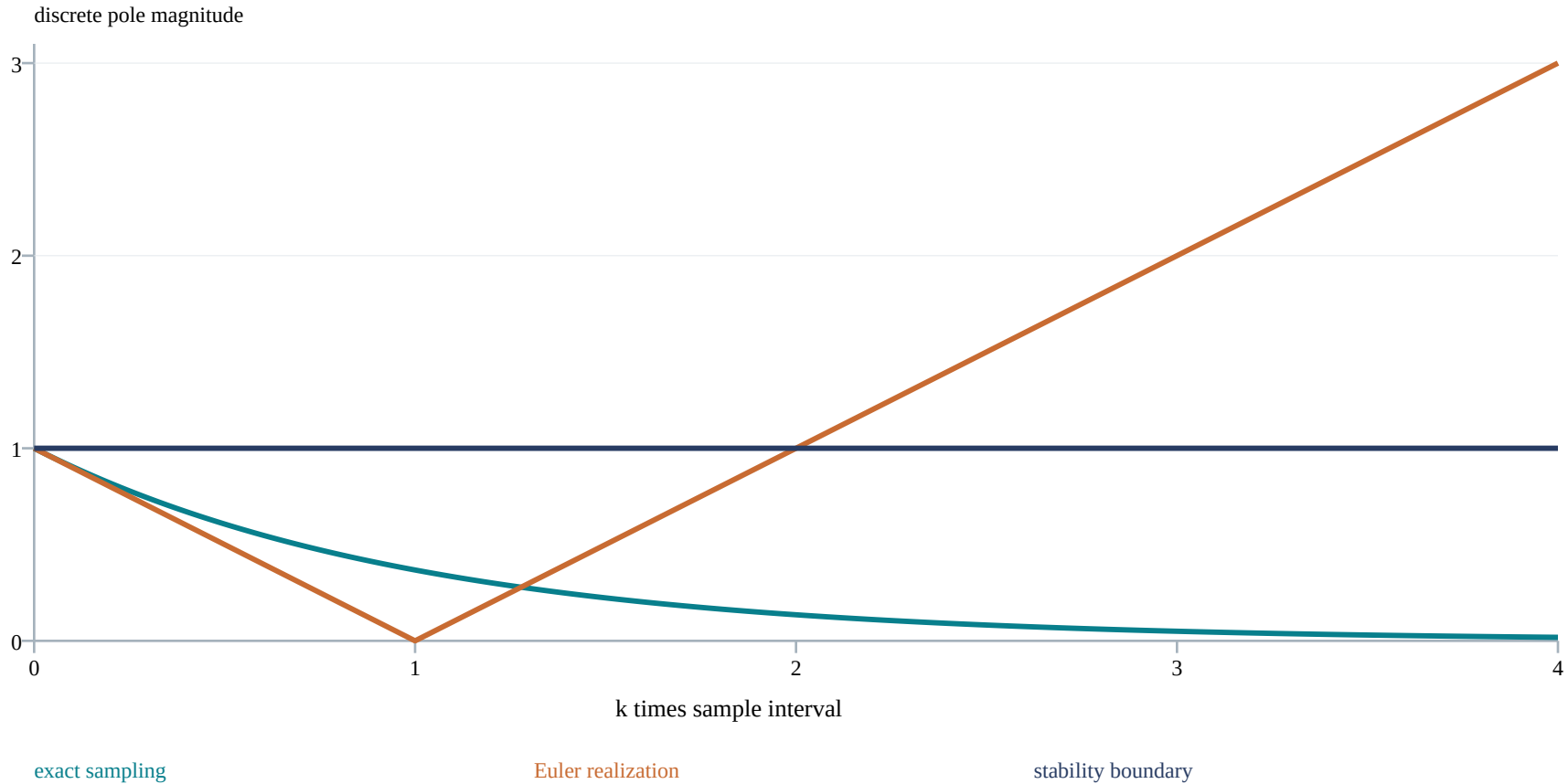
The update rule changes stability

Analytic scalar system; exact sampling versus forward Euler